

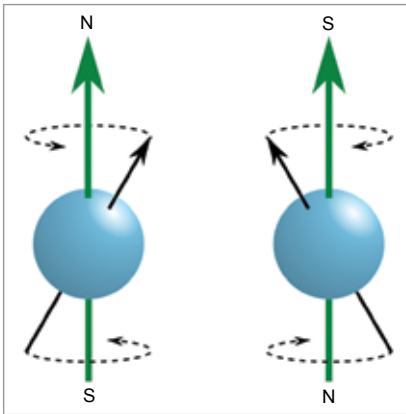


Fig. 2: Spinning electrons with associated magnetism

and process the information in the same location. Due to this fact the challenge of speed as well as power consumption are taken care of. Hence, the use of memristors in our conventional computing is paving a new way.

The property of memristors to store and process the information at the same location is like that of human brain's building blocks—neurons. With this capability, there is a paradigm shift in computing, called neuromorphic computing, which will mimic our internal neuron structure and energy-efficient processing like our brain. This will lead to putting a brain to a chip, that is, embedded artificial intelligence in the hardware itself, doing away with the need to run it on a separate software platform.

In the context of artificial intelligence (AI), neuromorphic computing by memristors will lead to a new generation of chips, which will be able to do the self-learning and interpretate the environment around it, without the help of coded software. Currently, the AI runs on cloud. With this neuromorphic computing, the AI running in central servers on cloud will shift to chip itself as edge computing.

In practical terms, an autonomous car requires network connectivity (4G/5G) through which

it sends the data collected through its sensors to its central server on the cloud for analysis. Central server does the data analysis and sends the execution command back to the car. In this process latency is created and it is also very power consuming to handle such a huge data. With the advent of memristors, all computations can be done locally inside the brain of the car. This will take the AI to a different level.

Neuromorphic computing will establish a new level in AI in edge computing, robotics, the Internet of Things (IoT), machine vision, artificial skin, sensory organ, and so on.

## Spintronics

Spintronics is a word coined from the words spin and electronics. The invention of this can be traced back to early 1988, when the giant magnetoresistive (GMR) effect was discovered by Albert Fert in France and Peter Gruenberg in Germany. Our conventional electronics harnesses the property of charge on the electrons, whereas spintronics harnesses the property of spinning of the electrons.

Conventional electronics and semiconductors define the state of a device by zero and one states. For example, a capacitor is said to be in 'one' state when it is fully charged and in 'zero' state when fully discharged. This is how our data is stored in '0' and '1' format in our memory devices with the help of billions of capacitors.

Apart from the charge on the electron, every electron spins on its axis in two different directions—clockwise and counter-clockwise. This motion is inherent to electrons and can be altered from one to another with some application. This spinning of electrons generates magnetic moment that can be altered with the application of external magnetic field. Spintronics uses this

property of electrons where clockwise rotating electrons are said to be in '0' state and counter-clockwise in '1' state.

In conventional way, charging and discharging of capacitors is power consuming. In order to maintain the state of a capacitor in '1' state, it has to be charged several times in a second, whereas the spin of an electron is inherent and it does not require any power to maintain the same. The advantage of this is that it is non-volatile, which means the memory does not go-off when the power is shutdown, which is the problem in conventional charge based electronics.

Spintronics is the driving technology for next-generation nano technology. It will enhance the memory capability and processing speed while reducing the power consumption. Quantum-mechanical computing based on spintronics can achieve a speed unheard of with conventional electrical computing.

Spinning of electrons, which is the fundamental of spintronics, can be managed in several ways, such as by applying:

- Magnetic field
- Stray electric and magnetic field
- Electromagnetic wave, for example, circularly polarised light
- A thermal gradient

While spintronics is a magnetic phenomenon of electrons, ferro magnetic material started as a core to this and different devices like giant-magneto-resistance, tunneling-magneto resistance, and magnetic-tunnel-junction are built around it. Despite paced advancement in ferrous material, semiconductor spintronics is also the part of focus to find a way to generate and control the spins of the electrons.

The selection of material for semiconductor spintronics depends on the property of material exhibiting ferromagnetism at room temper-